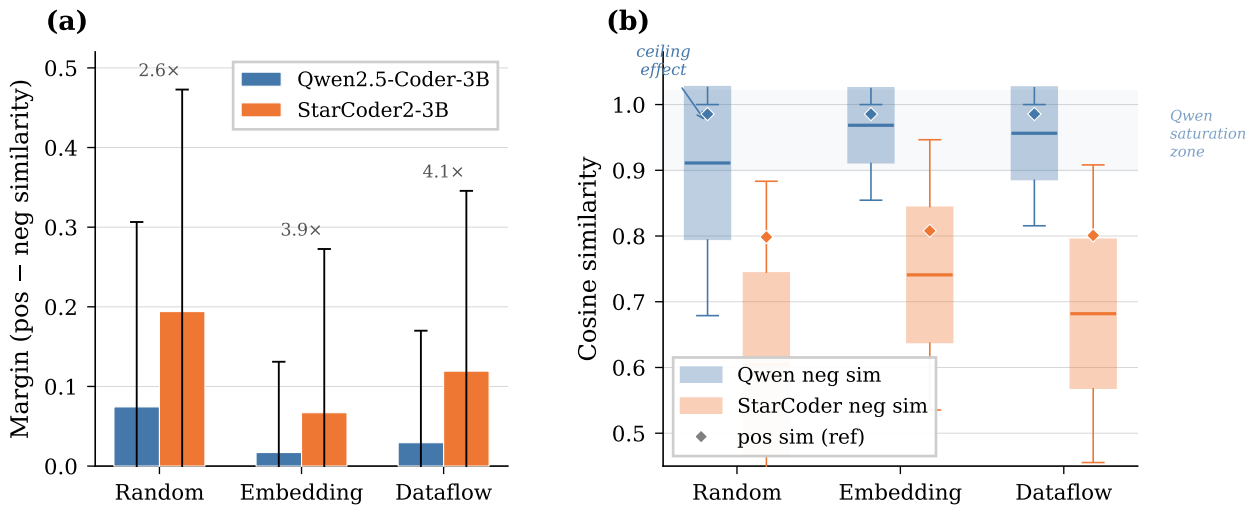
(a)
Margin (pos − neg similarity)
0.0
0.1
0.2
0.3
0.4
0.5
2.6×
3.9×
4.1×
Qwen2.5-Coder-3B
StarCoder2-3B
Random
Embedding
Dataflow
(b)
Cosine similarity
0.5
0.6
0.7
0.8
0.9
1.0
ceiling effect
Qwen saturation zone
Qwen neg sim
StarCoder neg sim
pos sim (ref)
Random
Embedding
Dataflow